



Description and genomic insights into a multidrug resistant novel bacterium *Savagea serpentis* sp. nov., isolated from the scats of a vine snake (*Ahaetulla nasuta*)

Vikas Sudhakar Patil · Yogita Lugani · Ravindra Deoram Chaudhari ·
Prachi Pramod Karodi · Pramod Chandrakant Mane · Priyam Mehrotra ·
Shrikant Prakash Pawar · Yogesh Shripad Shouche · Venkata Ramana Vemuluri 

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Abstract Two Gram-stain positive, endospore forming, non-motile, rod shaped bacterial strains SN6^T and SN6b were isolated from scats of a mildly venomous vine snake (*Ahaetulla nasuta*). Strains were phenotypically resistant to multiple antibiotics of four different classes i.e. aminoglycosides, β -lactams, fluoroquinolones and sulphonamides. Cells of both the strains were catalase positive and oxidase negative. Phylogenetic analysis based on 16S rRNA gene sequence analysis of these two strains showed closest similarity (99.2% and 99.3%) with *Savagea faecisuis* Con12^T, the only species of the genus *Savagea* and $\leq 94.9\%$ with the

species of other closest genera of the family *Planococcaceae*. The 16S rRNA gene sequence similarity (99%), DNA-DNA relatedness (95%) and similar phenotypic characteristics between the strains SN6^T and SN6b revealed their phylogenetic affiliation to the same species. Hence, strain SN6b is an additional strain of the type strain SN6^T. DNA-DNA relatedness of strain SN6^T with *S. faecisuis* Con12^T was 32.8%. Predominant fatty acids were iso-C_{15:0} (32.0%), iso-C_{16:1} ω 11c (19.2%) and iso-C_{17:1} ω 10c (12.1%). MK-6 (100%) was the only respiratory quinone of strain SN6^T. Diphosphatidylglycerol, phosphatidylglycerol and phosphatidylethanolamine were the major polar lipids. Cell wall peptidoglycan was A4 α ; L-Lys-Gly-D-Glu type. The DNA G + C content (mol%) of SN6^T was 40.8. Whole genome sequence of SN6^T consisted of 26,37,389 base pairs in length with 2667 annotated genes, out of which 1021 corresponds to hypothetical proteins and 1646 with functional

Vikas Sudhakar Patil and Yogita Lugani have contributed equally to this work.

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V. S. Patil · P. P. Karodi · S. P. Pawar (✉) ·
Y. S. Shouche · V. R. Vemuluri
National Centre for Microbial Resource, National Centre
for Cell Science, Pune, Maharashtra 411007, India
e-mail: pawar.shree@gmail.com

Present Address:
Y. Lugani · P. Mehrotra · V. R. Vemuluri (✉)
Microbial Type Culture Collection and Gene Bank, CSIR-
Institute of Microbial Technology, Chandigarh 160036,
India
e-mail: venkat@imtech.res.in

R. D. Chaudhari · P. C. Mane
P. G. Department of Zoology and Research Centre, Shri
Shiv Chhatrapati College, Junnar, Pune,
Maharashtra 410 502, India

P. Mehrotra · V. R. Vemuluri
Academy of Scientific and Innovative Research (AcSIR),
Ghaziabad 201002, India

assignments including antibiotic resistance, multidrug resistance efflux pumps, invasion and virulence factors. Comparative polyphasic study of the strains SN6^T, SN6b and *S. faecisuis* Con12^T elucidated the differentiating characteristics which led to describing strain SN6^T and SN6b as a novel species of the genus *Savagea* for which the name *Savagea serpentis* sp. nov. is proposed. The type strain of *Savagea serpentis* is SN6^T (= KCTC 33546^T = CCUG 6786^T).

Keywords *Ahaetulla nasuta* · Multidrug resistance · Phylogenetic analysis · Polyphasic taxonomic approach · *Savagea serpentis*

Introduction

In an attempt to study the bacterial diversity of gut of a mildly venomous snake (*Ahaetulla nasuta*), we isolated multidrug resistant strains SN6^T and SN6b which were showing closest relationship with *S. faecisuis* Con12^T. These two isolates were further subjected to polyphasic characterisation, which revealed their belongingness to the genus *Savagea*, and differentiating characteristics to their closest neighbour *S. faecisuis* Con12^T. The genomic insights of the strain SN6^T also revealed interesting details of multidrug resistance genes and *Mycobacterium* virulence operons (Gao et al. 2006), presumably obtained from horizontal gene transfer from the closest communities. At the time of writing, the genus *Savagea* is comprised of only one validly published type species *S. faecisuis* Con12^T (Parte et al. 2020; Whitehead et al. 2015). The genus *Savagea* has been delineated from other genera of the family *Planococcaceae*, based on its phenotypic, phylogenetic and chemotaxonomic differentiating characteristics. The current study led to the description of strain SN6^T as a novel species of the genus *Savagea*, for which, the name *Savagea serpentis* sp. nov. was proposed.

Materials and methods

Two strains SN6^T and SN6b were isolated from the scats of mildly venomous vine snake (*A. nasuta*). One gm of snake scats was dissolved into 100 ml of sterile Tryptone Soya Broth (TSB, HiMedia) adjusted to

different pH (range 3.0–7.0; with an interval of 1.0 pH unit) and incubated under shaking conditions (150 rpm) for 48 h. The average pH in most of the snakes' stomach after feeding reaches to a range of 3.0–4.0 from pH 7.0 due to secretion of gastric juices (Skoczylas 1970). Therefore, media with lower pH was also used to facilitate the favourable conditions during enrichment and isolation of bacterial strains. Turbid broth (50 and 100 µL) was spread onto two sets of Tryptone Soya Agar (TSA) plates of similar pH range mentioned above, followed by incubation at 37 °C for 48 h. Individual bacterial colonies with different morphology were separated by further streaking onto the TSA (HiMedia) plates. Strains were preserved in 15% (v/v) glycerol stocks at – 80 °C, in liquid nitrogen (LN2), and in lyophilized form. *S. faecisuis* Con12^T, the only type strain of the genus *Savagea*, was obtained from T. R. Whitehead, Bioenergy Research Unit, National Center for Agricultural Utilization Research, USDA, Agricultural Research Service, 1815 N. University Street, Peoria, IL 61,604, USA, for comparative analysis. Colony morphology and cell morphology of the strains SN6^T and SN6b, grown at 37 °C for 24 h, was observed using dissecting microscope and phase contrast microscope (Olympus-BX53F), respectively. Gram staining and spore staining were carried out using commercial kits (HiMedia) according to manufacturer's instructions. Cell motility was determined by hanging drop method as well as by the semi-solid agar method (Harley and Prescott 2002). TSA/TSB grown cells were used for physiological and biochemical characterisation unless otherwise specified. Assimilation of different organic compounds, enzyme production, nitrate reduction and acid production from various carbohydrates were tested by using API 20 NE, API ZYM and API 50CH strips (bioMérieux) according to manufacturer's instructions. Additional biochemical tests, carbon source utilization and enzymatic activities have been carried out on Gram-positive (GP) card by using "Vitek 2 compact" bioMérieux automated system (Wallet et al. 2005). Additionally, the cultures were screened for the production of different enzymes i.e. xylanase, amylase, pectinase and inulinase using the qualitative method. The cultures were streaked onto agar plates containing starch, pectin, and inulin, and incubated at 37 °C for 48 h in order to measure their amylase (Pinjari and Kotari 2018), pectinase (Oumer and Abate 2018) and inulinase (Navraj et al.

2016) activity. After incubation, the plates were flooded with iodine solution to check colonies showing a clear zone of substrate hydrolysis. The cultures were assayed for xylanase activity by spot plating the cultures on xylan containing agar plates followed by incubation at 37 °C for 48 h. The plates were further flooded with Congo red solution (0.5%, w/v) for 15 min and destained with NaCl (1.0 M) for 15 min. The bacterial colonies with a zone of inhibition which represented for xylanase activity were noted (Varghese et al. 2017). Gas production from D-glucose fermentation was tested by using Durham's tubes. IMViC tests were conducted for the confirmation of indole production, acid production by mixed acid fermentation of glucose, acetoin production and citrate utilization as described by Patil et al. (2015). H₂S production was tested on Triple Sugar Iron Agar and the results were interpreted by the absence of black precipitate. Catalase and oxidase tests were performed by using 3% (v/v) aqueous hydrogen peroxide solution and 1% (v/v) oxidase test reagent *N,N,N',N'*-tetramethyl-*p*-phenylenediamine dihydrochloride (HiMedia), respectively. Growth at several concentrations of NaCl (0–15%, w/v; with intervals of 1.0%), different pH (pH 4.0–11.0; with intervals of 1.0 pH unit) and temperature (5–55 °C; with intervals of 5 °C) was examined. Anaerobic growth of the strains was tested by inoculating them onto agar slants plugged with butyl Suba-Seal rubber septa. The air in the tubes was replaced by flushing in the inert gas argon using sterile syringe needles and incubated at 37 °C for 48 h. All the physiological tests were carried out in triplicates and the bacterial growth (OD₆₀₀) in the broth culture was measured using a spectrophotometer. Antibiotic susceptibility test was done by disc diffusion method on TSA plate by using filter paper antibiotic discs (6 mm diameter, HiMedia). Susceptibility was determined by the zone of inhibition. For 16S rRNA and *gyrB* (encodes for DNA gyrase subunit-B) gene sequencing, DNA was extracted from log phase cultures by phenol–chloroform method (Marmur 1961) with additional *RNase* treatment. The 16S rRNA and *gyrB* genes were amplified (Sambrook et al. 1989) by using eubacterial universal primers 27F (5'-AGAGTTTGATCMTGGCTCAG-3'), 1492R (5'-ACGGCTACCTTGTTACGACTT3') [11] and *gyrB*BAUP2 (5'-GCGGAAGCGGCCNGS-NATGTA-3'), *gyrB*Bndn1 (5'-CCGTCCACGTCGGCRTCNGYCAT-3'),

respectively. Amplified products were purified by polyethylene glycol (PEG)–NaCl method (Lane 1991). DNA strands of the amplicons were sequenced in an ABI 3730 xl DNA analyzer using the Big Dye terminator kit (Applied Biosystems). The sequences thus obtained were assembled using DNA STAR Navigator (version 15.2.0.130) followed by sequence similarity analysis using online databases EZBIO-CLOUD (Yoon et al. 2017) and National Center for Biotechnology Information (NCBI). The 16S rRNA and *gyrB* gene sequences of closely related members of the genus *Savanea* and members of closest genera were retrieved from NCBI, and were aligned along with the sequences of strains SN6^T and SN6b using Clustal-W tool of MEGA7 software. The best fit model for phylogenetic analysis was identified through “Find best DNA/protein model” tool of MEGA7. Phylogenetic trees were constructed by using neighbor joining (NJ) method with Kimura-2-parameter as a model of nucleotide substitution and 1000 bootstrap replications with Gamma distribution (Kumar et al. 2016). To confirm the genomic variation between strains SN6^T and *S. faecisuis* Con12^T, and SN6^T and SN6b, DNA–DNA hybridization (DDH) has been carried out in triplicates as described by Loveland-Curtze et al. (2011). For DDH and G + C content (mol%) analysis, DNA was isolated by phenol/chloroform method (Marmur 1961) and suspended in the 2X SSC (Sodium Saline Citrate) buffer. DNA was then subjected to ultrasonic shearing to get fragments of uniform size which was used to make homoduplex and heteroduplex DNA. The resulted DNA was mixed with SYBR Green-I (Invitrogen), a fluorescent dye specifically binds to double stranded DNA (dsDNA). The relative rate of reassociation of homoduplex and heteroduplex DNA was analysed by denaturation followed by optimum reassociation (Gillis et al. 1970) by using StepOnePlus Real-Time PCR system (Applied Biosystems). DNA–DNA relatedness in terms of relative binding ratio was calculated as described elsewhere (De Ley et al. 1970).

The whole genome sequencing of strain SN6^T was carried out on an Illumina MiSeq sequencing platform using paired end 2 × 150 base pair (bp) chemistry. The resultant reads were assembled by using PATRIC web server, version 3.6.7. (Davis et al. 2020). The genome annotation including the major subsystems and genomic DNA G + C content (mol%) analysis was performed using the RAST web server version 2.0

(Brettin et al. 2015) Antibiotic resistance determinants were identified by using web portal—RGI 5.1.1, CARD 3.1.0 (Alcock et al. 2020). CRISPR-CasFinder online tool (Couvin et al. 2018) was used to identify the CRISPR-Cas arrays and spacers with default parameters. AntiSMASH web tool version 5.2.0 (Blin et al. 2019) was used to identify the genes corresponds to secondary metabolite production in strain SN6^T.

To carry out whole cell fatty acid analysis, strains were inoculated onto TSA medium (pH 7.0) and incubated at 28 ± 2 °C for 18 h. Cells were harvested and then subjected to sequential processes of saponification, methylation, extraction and base wash. Resulted methyl esters of fatty acids were analysed by Gas Chromatography (Agilent Technologies; 7890 A) as per Microbial Identification System software (MIS, MIDI Inc., Newark, DE, USA; version 6.0). Peaks representing the fatty acids were identified based on the RTSBA6 database (Sasser 1990). Analysis of respiratory quinones and peptidoglycans was outsourced to DSMZ, Braunschweig, Germany, which was carried out as described by Tindall (1999a, b) and Schumann (2011), respectively. Polar lipids were extracted from freeze-dried culture in chloroform/methanol/0.3% (w/v) saline (1:2:0.8, v/v) as described by Bligh and Dyer (1959) with modifications by Card (1973). Lipids were separated on silica gel TLC (Kieselgel 60 F254; Merck) by two-dimensional chromatography using chloroform/methanol/water (65:25:4, v/v) for the first dimension and chloroform/methanol/acetic acid/water (80:12:15:4, v/v) for second dimension (Tindall 1990b). Air dried plates were stained with 5% (w/v) ethanolic molybdophosphoric acid for total lipids. Lipid functional groups were identified by using spray reagents ninhydrin (for amino groups), molybdenum blue (phosphates), Dragendorff (choline group) and α-naphthol (sugars).

Results and discussion

Strains SN6^T and SN6b grew well and appeared off-white in colour on TSA plate. Cells were non-motile rods (0.8–1.0 µm × 2.3 – 5.9 µm; Supplementary Fig. 1) and Gram-stain positive, whereas, cells of *S. faecisuis* Con12^T were motile. Though the endospore formation was reported to be absent in the type strain *S. faecisuis* Con12^T (Whitehead et al. 2015), terminal endospores were observed in the cells

of *S. faecisuis* Con12^T as well as in SN6^T and SN6b. Weak anaerobic growth was observed with the novel strain, which was not found in *S. faecisuis* Con12^T. Both the strains were oxidase negative and catalase positive, whereas, *S. faecisuis* Con12^T was negative for catalase and weakly positive for oxidase. Strain SN6^T could utilize only glycerol, D-ribose, arbutin, esculin, D-trehalose, gentiobiose, gluconate, 2-keto-gluconate and 5-keto-gluconate amongst 49 carbohydrates tested (API 50CH). Strain SN6b also showed similar pattern of carbohydrate assimilation except for D-ribose and esculin which showed weakly positive reactions, in contrast to *S. faecisuis* Con12^T, which was found to be negative for the utilization of D-ribose, arbutin, esculin, D-trehalose, gentiobiose, gluconate and 2-keto-gluconate. Acid production was observed with glycerol, D-ribose, arbutin, gentiobiose and 2-ketogluconate out of 49 carbohydrates tested, but not observed in *S. faecisuis* Con12^T. Gas production from D-glucose was absent in the strain SN6^T. Hydrolysis of urea, gelatin, L-arginine, β-D galactopyranoside was not detected, whereas, *S. faecisuis* Con12^T could hydrolyze gelatin. Nitrate reduction, indole production, H₂S production, acid production from D-glucose, and butylene glycol pathway was found to be absent in SN6^T (SN6b). Strains showed positive activity for five enzymes i.e. esterase (C4), esterase lipase (C8), naphthol-AS-BI-phosphohydrolase, alkaline and acid phosphatase out of twenty different enzymes tested (API ZYM), whereas, *S. faecisuis* was weakly positive for esterase (C4), esterase lipase (C8), leucine arylamidase and negative for naphthol-AS-BI-phosphohydrolase, and alkaline and acid phosphatase. Strains SN6^T and SN6b were positive for inulinase and negative for amylase, pectinase, and xylanase, whereas, *S. faecisuis* Con12^T was positive for amylase and weakly positive for inulinase, pectinase, and xylanase. The optimum temperature of growth for strains SN6^T and SN6b was 40 °C (range 20–45 °C), pH 8.0 (7.0–10.0) and NaCl 3.0% (w/v), whereas, *S. faecisuis* Con12^T showed adequate growth at an optimum temperature of 30 °C and did not grow at 45 °C. Novel strains did not require NaCl for their growth, however, they could tolerate up to 7.0% (w/v), while, *S. faecisuis* Con12^T could tolerate up to 9.0% NaCl. Novel strains were observed to be resistant to multiple antibiotics (µg per disc unless indicated) belonging to four different classes i.e. streptomycin-10

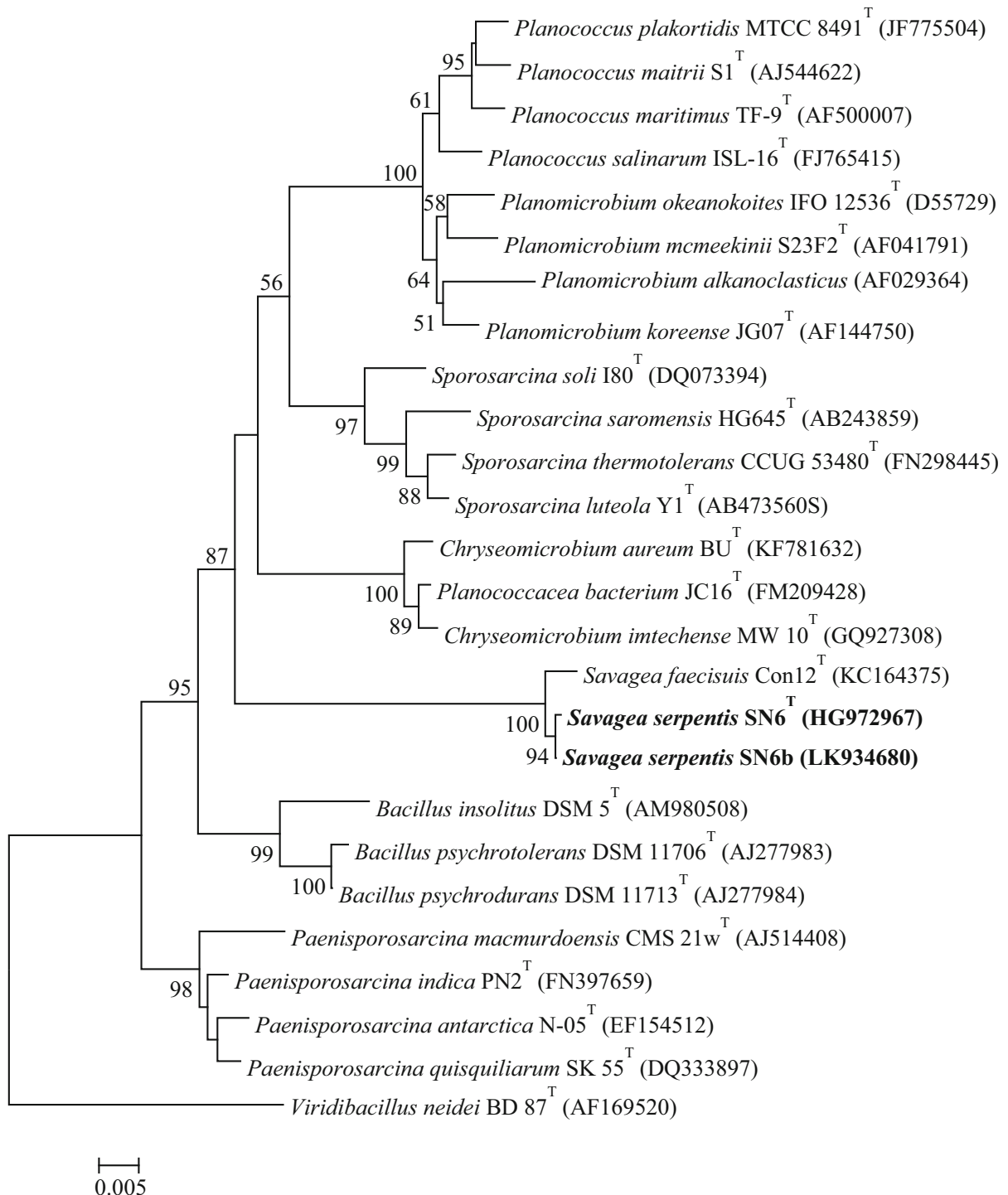


Fig. 1 A neighbor-joining phylogenetic tree based on 16S rRNA gene sequences depicting the phylogenetic relatedness of strains SN6^T and SN6b along with closest type strain *S. faecisuis* Con12^T, and members of the closely related genera of

Planococcaceae family. Bootstrap values (> 50%) obtained from 1000 replicates are expressed as percentages at branching nodes. *Viridibacillus neidei* BD-87^T (AF169520) is included as an out group. Bar, 0.005 substitutions per site

(Class-aminoglycoside); cefpodoxime-10 (β -lactam); norfloxacin-10, ofloxacin-5 (class-fluoroquinolone) co-trimoxazole-25, sulphatriad-300 (class-sulphonamide), however, susceptible to ampicillin-10, amoxycillin-10, amikacin-30, augmentin-30, cefotaxime-30, cefixime-5, cefuroxime-30, ceftriaxone-30, cephalothin-30, cloxacillin-5, colistin-10, furazolidone-50, levofloxacin-5, oxacillin-1, penicillin-G (1 unit), ciprofloxacin-5, rifampicin-5, vancomycin-30, clindamycin-2, lincomycin-2, erythromycin-15, chloramphenicol-25, doxycycline hydrochloride-30, nitrofurantoin-300, tetracyclin-25, fosfomycin-200, whereas, *S. faecisuis* Con12^T was resistant to ofloxacin, co-trimoxazole and sulphatriad, which belong to two different classes of antibiotics. Both the strains SN6^T and SN6b showed the closest similarity (99.2% and 99.3%) to *S. faecisuis* Con12^T, the only species of the genus *Savagea* and $\leq 94.9\%$ similar to all the species of other genera of the family *Planococcaceae*. The strains SN6^T and SN6b were 99% similar based on 16S rRNA gene analysis and share similar phenotypic characteristics in addition to phylogenetic clustering. Based on the polyphasic similarity between SN6^T and SN6b, we proposed SN6^T as type strain and SN6b as an additional strain (of types strain). The genomic relatedness between the novel strain SN6^T and *S. faecisuis* Con12^T was analysed by the DNA-DNA hybridization (DDH) which was found to be 32.8%, which was significantly lower than the threshold value for the delineation of SN6^T as a novel species (Stackebrandt and Ebers 2006). The DNA relatedness between strains SN6^T and SN6b was 95.0%, which proved that these two strains belong to the same species (Wayne et al. 1987). Phylogenetic analysis based on 16S rRNA gene and protein coding gene sequence *gyrB* revealed that strains SN6^T and SN6b formed a distant clade with *S. faecisuis* Con12^T and then with other the members of various genera of *Planococcaceae* family (Fig. 1, Supplementary Fig. 2). The *gyrB* gene sequence of SN6^T and SN6b showed closest similarity of 81.92% and 82.04%, respectively, to the closest relative of the genus *S. faecisuis* Con12^T. The *gyrB* gene sequence similarity between SN6^T and SN6b was 99%. The genomic DNA G + C content (mol%) of strain SN6^T was 40.8 which varied with that of the closest type strain *S. faecisuis* Con12^T (41.8).

The GenBank accession number of the genome sequence of strain SN6^T is JADKPV000000000.

Unicycler based de novo genome assembly of strain SN6^T yielded 21 contigs. The genome sequence of strain SN6^T consisted of 26,37,389 bp. The genome annotation resulted in the assignment of 2,667 genes, out of which 1,646 with functional assignments and 1,021 corresponds to hypothetical proteins. In addition to the exhibited phenotypic resistance to antibiotics of the classes β -lactams, fluoroquinolone and sulphonamides, strain SN6^T also contained several antibiotic resistance genes such as tet(45) codes for tetracycline resistance, MFS efflux pump; lnuG for lincosamide nucleotidyltransferase, which facilitates lincosamide antibiotic inactivation and APH(2'')-IIIa for aminoglycoside antibiotic inactivation, represented the type strain as an emerging extensive drug resistant bacterium. Presence of *Mycobacterium* virulence operons (essential for virulence) and, invasion and intracellular resistance in strain SN6^T determined its ability to invade and grow inside a host cell, which is also an important virulence property of pathogenic *Mycobacteria* (Gao et al. 2006). Snakes were known to contain several acquired drug resistant bacteria in their gut, which might be transferred to them through the food chain (Laxminarayan et al. 2013). During the flow of antibiotics from non-living habitats to living things, the microbial communities present in respective hosts are presumably emerging and acquiring the drug resistance and virulence factors through horizontal gene transfer (Bonnedahl et al. 2009). In the similar pattern, strain SN6^T might have acquired multiple drug resistance and virulence determinants from their closest communities. Though the type strain is phenotypically resistant to three different classes of antibiotics, its genome reveals also the occurrence of antibiotic resistance genes corresponding to the classes of tetracyclines, aminoglycosides and lincosamides. In this case, strain SN6^T might have acquired the partial genome components of antibiotic resistance against tetracyclines, aminoglycosides and lincosamides by horizontal gene transfer, which currently might not be allowing the strains to be functionally resistant against antibiotic clindamycin. However, the strain may likely to become functionally resistant against tetracyclines, aminoglycosides and lincosamides after acquiring necessary additional genes for the expression of antibiotic resistance (Jaillard et al. 2017) in near future. Since, pathogenicity was not tested phenotypically in the current study, the authors did not confirm the pathogenicity of the

strain SN6^T due to the presence of such virulent determinants. However, the occurrence of such virulence factor hitherto unexplored in the validated *Sav-
agea* species given a clue for the predictable horizontal gene transfer from the microbial populations of the micro environment. The functional annotation of genes of strain SN6^T to several major metabolic subsystems was given in supplementary Fig. 3. Genome sequence of strain SN6^T has 30 CRISPR-Cas spacers distributed among seven different CRISPR arrays. Presence of diverse CRISPR-Cas spacers predicts the extent of defence mechanism developed by the type strain against viral genomes. Genes corresponding to secondary metabolite (betalactone) were observed in the genome sequence of strain SN6^T. The gene cluster of these metabolites include core genes, additional biosynthetic genes and regulatory genes but did not have transport related genes. Occurrence of genes corresponds to secondary metabolite may be a tip to exploit strain SN6^T for phenotypic expression of the betalactone in near future.

Major cellular fatty acids of strains were iso-C_{15:0} (32%), C_{16:1} ω11c (19.2%) and iso-C_{17:1} ω10c (12.1%) with minor amounts of C_{16:0}, iso-C_{16:0}, iso-C_{17:0}, anteiso-C_{15:0}, C_{17:1} ω9c, C_{16:1} ω7c alc and anteiso-C_{17:0} (Table 1). There was a noticeable quantitative variation between the fatty acids reported (Whitehead et al. 2015) and results obtained from the current study of *S. faecisuis* Con12^T. Though there was a close resemblance of the qualitative fatty acid profile of strain SN6^T and *S. faecisuis* Con12^T, we noticed a minor quantitative difference among them. Respiratory quinone of type strain SN6^T was MK6 (100%) which was similar to *S. faecisuis* Con12^T. Cell wall peptidoglycan analysis of type strain SN6^T belonged to A4α type (sub type A11.40) represented by the presence of cross-linkage L-Lys-Gly-D-Glu. Both of these characteristics were similar to that of *S. faecisuis* Con12^T, which supported the affiliation of SN6^T to the genus *Sav-
agea*. Polar lipids of the strain SN6^T comprised of phosphatidylglycerol (PG), diphosphatidylglycerol (DPG), phosphatidylethanolamine (PE), unidentified phospholipid (UPL), aminophospholipid (APL) and an unidentified lipids (UL) (Supplementary Fig. 4). Furthermore, novel strains of the present study SN6^T and SN6b showed significant phenotypic and genotypic differences with its closest phylogenetic neighbour *S.*

Table 1 Fatty acid profile of strain SN6^T and *S. faecisuis* Con12^T (Data from current study)

Fatty acids*	SN6 ^T	<i>S. faecisuis</i> Con12 ^T
<i>Saturated</i>		
C _{14:0}	5.6	6.0
C _{16:0}	5.7	5.5
C _{16:0} N alc	1.5	2.3
<i>Unsaturated</i>		
C _{16:1} ω7c alc	2.2	2.5
C _{16:1} ω11c	19.2	25.4
C _{17:1} ω9c	2.4	2.0
<i>Branched</i>		
i-C _{14:0}	1.9	1.5
i-C _{15:0}	32	34.8
ai-C _{15:0}	4.9	3.5
i-C _{16:0}	1.3	0.9
i-C _{17:0}	5.6	3.0
i-C _{17:1} ω10c	12.1	8.3
ai-C _{17:0}	1.8	1.1
Sum-In-Feature 4 ^E	1.2	1.2

i, iso-branched; ai, anteiso-branched

^Ei-C_{17:1} I/ai-B, *fatty acids of > 0.9% were given in the table

Major fatty acids (> 10%) were bold

faecisuis Con12^T which indicated their distant relatedness (Table 2, Fig. 1). Based on the extent of 16S rRNA gene sequence similarity, genus level phylogenetic clustering and genus specific chemotaxonomic characteristics, strain SN6^T was assigned to the genus *Sav-
agea*. However, based on the significant differences in morphological, physiological, molecular, biochemical and phylogenetic characteristics (Table 2) from its closely related type strain *S. faecisuis* Con12^T, strain SN6^T is represented as a novel species of the genus *Sav-
agea*, for which the name *Sav-
agea serpentis* sp. nov. is proposed.

Description of *Sav-
agea serpentis* sp. nov.

*Sav-
agea serpentis* (ser.pen'tis. L. gen. n. serpentis, of a snake, referring to the source of isolation of type strain).

Cells are Gram-stain-positive, non-motile rods and form endospores. Facultative anaerobe, catalase positive and oxidase negative. Growth occurs at a range of

Table 2 Differentiating characteristics between strain SN6^T (*n* = 2) and *S. faecisuis* Con12^T

Characteristics	SN6 ^T (SN6b)	<i>S. faecisuis</i> Con12 ^T
Cell shape	Rods	Rods/Spherical
Motility	—	+
Temp optimum (°C)	40	30
Growth at 45 °C	+	—
NaCl range (% w/v)	0–7.0	0–9.0
Anaerobic growth	+	—
Catalase	+	—
Oxidase	+	(+)
Acetoin production	—	+
<i>Hydrolysis of</i>		
Gelatin	—	+
<i>Acid production</i>		
Glycerol	+	—
D-ribose	+	—
Arbutin	+	—
Gentiobiose	+	—
2-ketogluconate	+	—
<i>Carbohydrates utilized</i>		
Glycerol	+	—
D-ribose	+	—
Arbutin	+	—
Esculin	+	—
D-trehalose	+	—
β-gentiobiose	+	—
Gluconate	+	—
2-ketogluconate	+	—
<i>Enzyme production</i>		
Esterase (C4)	+	(+)
Esterase lipase (C8)	+	(+)
Leucine arylamidase	—	(+)
Acid phosphatase	+	—
Naphthol-AS-BI-phosphohydrolase	+	—
Alkaline phosphatase	+	—
Xylanase	—	(+)
Amylase	—	+
Pectinase	—	(+)
DNA G + C content (mol%)	40.8	41.8

+, positive; —, negative; (+), weak positive

Data given in the table was obtained from current study except DNA G + C (mol%) for strain Con12^T which was taken from literature (Whitehead et al. 2015)

temperature 20–45 °C (optimum 40 °C), pH 7.0–10.0 (optimum 8.0) and NaCl 0–7.0% (optimum 3.0%). Type strain SN6^T utilizes glycerol, D-ribose, arbutin, esculin, D-trehalose, gentiobiose, gluconate, 2-ketogluconate and 5-keto-gluconate among 49 carbohydrates tested by API-50 CH. Produces acid from glycerol, D-ribose, arbutin, gentiobiose, and

2-ketogluconate. Do not hydrolyze urea, gelatin, L-arginine, β-D galactopyranoside and weak positive for esculin. Negative for nitrate reduction, indole production, H₂S production, acid production from D-glucose and Voges-Proskauer test. Positive for esterase (C4), esterase lipase (C8), naphthol-AS-BI-phosphohydrolase, alkaline, and acid phosphatase

among twenty different enzymes tested (API ZYM). Major cellular fatty acids are iso-C_{15:0} (32.09%), iso-C_{16:1} ω11c (19.29%) and iso-C_{17:1} ω10c (12.15%). Cell wall peptidoglycans are A4α-L-Lys-Gly-D-Glu (Type A11.40). MK 6 (100%) is the sole respiratory quinone. Polar lipids comprise of phosphatidylglycerol (PG), diphosphatidylglycerol (DPG), phosphatidylethanolamine (PE), unidentified phospholipid (UPL), aminophospholipid (APL) and an unidentified lipid (UL). The DNA G + C content (mol%) is 40.8. The type strain SN6^T (= KCTC 33546^T, = CCUG 6786^T) was isolated from the scats of a vine snake (*A. nasuta*).

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Author contributions Samples were collected by RDC and PCM. Work conceptualized by VRV, SPP and RDC. VSP and PPK isolated the strains. Characterisation and deposition of strains and whole genome sequence done by VSP and PPK, VRV, YL and PM. Manuscript preparation and revision by VRV and YL. Phylogenetic and genomic analysis done by VRV. The whole study was supervised by VRV, SPP and YSS. Funding and resources acquisition by YSS, VRV and SPP.

Data availability All the data included in this study are available with the corresponding author VRV. *Nucleotide sequence data*: The GenBank / EMBL / DDBJ accession numbers for the 16S rRNA, *gyrB* gene sequences of strains SN6^T and SN6b are HG972967, KT198043 and LK934680, KT198044, respectively. The accession number for whole genome sequence of strain SN6T is JADKPV000000000. *Strain repositories*: KCTC 33546^T = CCUG 6786^T.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethical approval Our study was done using the scat samples of a snake collected from the branch of a tree (in college campus) during snake freshly passed by. We neither handled, nor catch hold of snake to carry out our study. No In-vivo experiment was carried out on the animal. The ARRIVE guidelines

apply for In-vivo experiments done with animals. Hence, “ethical approval” is not applicable in our study.

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